

Expectation Maximization: From Mixture Densities to a Provably Convergent Algorithm

Richard Xu

Lecture notes and tutorial

Abstract

Data with several clusters cannot be described by a single Gaussian, and the natural remedy — a mixture of Gaussians — produces a log-likelihood that neither yields a closed-form maximum nor suits plain gradient ascent, since the objective is non-concave. These notes develop the Expectation-Maximization (E-M) algorithm as the standard answer. After reviewing convexity and Jensen’s inequality, we state the E-M update and prove its central guarantee: each iteration leaves the log-likelihood non-decreasing. The proof is given twice — once through the ELBO–KL decomposition familiar from variational inference, viewing E-M as maximization-maximization, and once directly through the classical Q/H decomposition, where a one-line Jensen argument shows the cross-entropy term can never decrease. The framework is then applied in full to the Gaussian Mixture Model: responsibilities form the E-step, and Lagrange multipliers plus matrix calculus give closed-form M-step updates for the mixing weights, means, and covariances, each a responsibility-weighted average. A sanity check confirms the updates collapse to the single-Gaussian maximum-likelihood estimates when $k = 1$. Exercises with worked solutions conclude the notes.

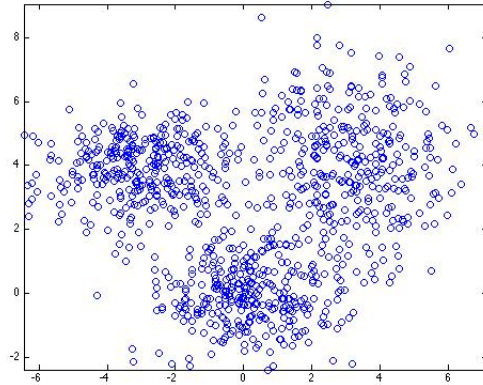
Keywords: expectation maximization; Gaussian mixture model; latent variables; Jensen’s inequality; ELBO; maximum likelihood

Contents

1	Motivation: Mixture Density Models	3
1.1	Gaussian Mixture model result	3
2	Preliminaries	4
2.1	Convex function	4
2.2	Jensen's inequality	4
2.2.1	Jensen's inequality example: $-\log(x)$	5
3	Expectation-Maximization Algorithm	5
4	Proof of convergence: Maximization-Maximization	6
4.1	Maximization of ELBO	6
4.1.1	why $H(\Theta \Theta^{(g)}) \geq H(\Theta^{(g)} \Theta^{(g)}) \quad \forall \Theta$	8
4.2	Proof of convergence without ELBO	8
5	E-M Example: Gaussian Mixture Model	9
5.1	Gaussian Mixture Model in action	9
5.2	The E-Step	9
5.3	Some derivation to help	9
5.3.1	Proof	10
5.4	The M-Step objective function	10
5.4.1	computing responsibilities	11
5.5	The M-Step: maximizing α	11
5.6	The M-Step: maximizing μ, Σ	12
5.6.1	Some formulas to remember	12
5.6.2	Maximization μ_l	13
5.6.3	Maximization of covariance	13
5.7	Summary of Gaussian Mixture Model	14
5.8	GMM fitting: initialization vs. convergence	15
6	Exercises	16
7	Solutions	16

1 Motivation: Mixture Density Models

When you have data that looks like:



Can you fit them using a single-mode Gaussian distribution, i.e.,:

$$\begin{aligned} p(x) &= \mathcal{N}(x|\mu, \Sigma) \\ &= (2\pi)^{-k/2} |\Sigma|^{-\frac{1}{2}} \exp\left(-\frac{1}{2}(x - \mu)^\top \Sigma^{-1}(x - \mu)\right) \end{aligned} \quad (1)$$

Clearly not! Data like this is typically modelled using mixture densities [1] — in this case, a Gaussian Mixture Model with k mixtures (GMM):

$$p(x) = \sum_{l=1}^k \alpha_l \mathcal{N}(x|\mu_l, \Sigma_l) \quad \text{s.t.} \quad \sum_{l=1}^k \alpha_l = 1 \quad (2)$$

1.1 Gaussian Mixture model result

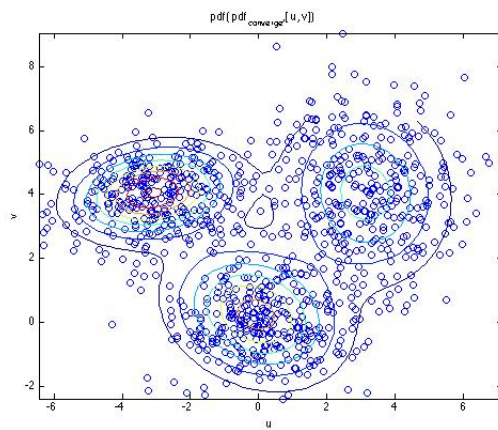


Figure 1: GMM fitting result

Let $\Theta = \{\alpha_1, \dots, \alpha_k, \mu_1, \dots, \mu_k, \Sigma_1, \dots, \Sigma_k\}$

$$\begin{aligned}
\Theta_{\text{MLE}} &= \arg \max_{\Theta} \mathcal{L}(\Theta|X) \\
&= \arg \max_{\Theta} \left(\sum_{i=1}^n \log \sum_{l=1}^k \alpha_l \mathcal{N}(x_i | \mu_l, \Sigma_l) \right) \quad \sum_{l=1}^k \alpha_l = 1
\end{aligned} \tag{3}$$

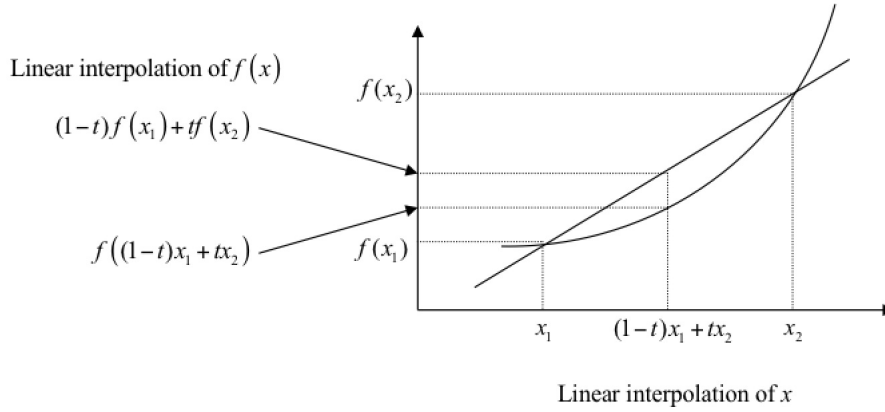
1. Unlike the single-mode Gaussian, we cannot simply take derivatives and set them to zero, i.e., optimize it analytically.
2. In terms of optimizing its parameters: the problem is **non-concave**. So traditional gradient ascent is **not** suitable for it, i.e., there are many local maxima.

The **goal** is to find an iterative process that ensures that $\log(p(X|\Theta^{(g)}))$ remains non-decreasing for $g = 1, \dots$

To do so, for data $X = \{x_1, \dots, x_n\}$, we introduce “latent” variables $Z = \{z_1, \dots, z_n\}$, each z_i indicating which mixture component x_i belongs to. The core problem then simply boils down to fitting a single Gaussian.

2 Preliminaries

2.1 Convex function



$$f((1-t)x_1 + tx_2) \leq (1-t)f(x_1) + tf(x_2) \quad t \in [0, 1] \tag{4}$$

2.2 Jensen's inequality

Using notation ϕ instead of f :

$$\phi((1-t)x_1 + tx_2) \leq (1-t)\phi(x_1) + t\phi(x_2) \quad t \in [0, 1] \tag{5}$$

Can be generalized further for any convex combination, i.e., let $\sum_{i=1}^n p_i = 1$:

$$\begin{aligned}
\phi(p_1x_1 + p_2x_2 + \dots + p_nx_n) &\leq p_1\phi(x_1) + p_2\phi(x_2) + \dots + p_n\phi(x_n) \quad \sum_{i=1}^n p_i = 1 \\
\implies \phi\left(\sum_{i=1}^n p_i x_i\right) &\leq \sum_{i=1}^n p_i \phi(x_i) \\
\implies \phi\left(\sum_{i=1}^n p_i f(x_i)\right) &\leq \sum_{i=1}^n p_i \phi(f(x_i)) \quad \text{by replacing } x_i \text{ with } f(x_i)
\end{aligned} \tag{6}$$

generalize to the continuous case:

$$\phi \left(\int_x f(x)p(x)dx \right) \leq \int_x \phi(f(x))p(x)dx \implies \phi(\mathbb{E}[f(x)]) \leq \mathbb{E}[\phi(f(x))] \quad (7)$$

2.2.1 Jensen's inequality example: $-\log(x)$

$\phi(x) = -\log(x)$ is a convex function:

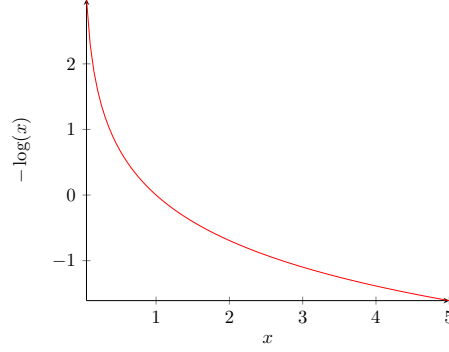


Figure 2: example of convex function $-\log(x)$

1. when $\phi(\cdot)$ is convex:

$$\begin{aligned} \phi(\mathbb{E}[f(\mathbf{x})]) &\leq \mathbb{E}[\phi(f(\mathbf{x}))] \\ \text{e.g. } -\log(\mathbb{E}[f(\mathbf{x})]) &\leq \mathbb{E}[-\log(f(\mathbf{x}))] \end{aligned} \quad (8)$$

2. when $\phi(\cdot)$ is concave:

$$\begin{aligned} \phi(\mathbb{E}[f(\mathbf{x})]) &\geq \mathbb{E}[\phi(f(\mathbf{x}))] \\ \text{e.g. } \log(\mathbb{E}[f(\mathbf{x})]) &\geq \mathbb{E}[\log(f(\mathbf{x}))] \end{aligned} \quad (9)$$

3 Expectation-Maximization Algorithm

Instead of performing:

$$\begin{aligned} \Theta^{\text{MLE}} &= \arg \max_{\Theta} \mathcal{L}(\Theta|X) \\ &= \arg \max_{\Theta} (\log[p(X|\Theta)]) \end{aligned} \quad (10)$$

The trick is to introduce some “latent” variable Z into the model.

For each iteration of the E-M algorithm [2], we perform:

$$\Theta^{(g+1)} = \arg \max_{\Theta} \left(\int_Z \log(p(X, Z|\Theta))p(Z|X, \Theta^{(g)})dZ \right) \quad (11)$$

However, before we apply it, we must ensure convergence [3], i.e.,

$$\begin{aligned}
\log(p(X|\Theta^{(g+1)})) &= \mathcal{L}(\Theta^{(g+1)}|X) \\
&\geq \mathcal{L}(\Theta^{(g)}|X) \\
&= \log(p(X|\Theta^{(g)})) \quad \forall g
\end{aligned} \tag{12}$$

Note the difference between the variable Θ and the constant $\Theta^{(g)}$. Also note that gradient ascent is **not** suitable for many E-M problems, as they can be **non-concave**.

4 Proof of convergence: Maximization-Maximization

We have seen this in the variational-inference literature [4], as the ELBO-KL decomposition:

$$\begin{aligned}
\mathcal{L}(\Theta|X) &= \log(p(X|\Theta)) \\
&= \log\left(\frac{p(X, Z|\Theta)}{p(Z|X, \Theta)}\right) \\
&= \log\left(\frac{p(X, Z|\Theta)}{q(Z)} \times \frac{q(Z)}{p(Z|X, \Theta)}\right) \\
&= \log\left(\frac{p(X, Z|\Theta)}{q(Z)}\right) + \log\left(\frac{q(Z)}{p(Z|X, \Theta)}\right) \\
&= \int_Z \log\left(\frac{p(X, Z|\Theta)}{q(Z)}\right) q(Z) dZ + \int_Z \log\left(\frac{q(Z)}{p(Z|X, \Theta)}\right) q(Z) dZ \\
&= \text{ELBO}(\Theta, q) + \text{KL}(q(Z)||p(Z|X, \Theta))
\end{aligned} \tag{13}$$

or to use Jensen's inequality:

$$\begin{aligned}
\mathcal{L}(\Theta|X) &= \log p(X|\Theta) = \log \int_Z p(X, Z|\Theta) dZ \\
&= \log\left(\int_Z \frac{p(X, Z|\Theta)}{q(Z)} q(Z) dZ\right) \\
&\geq \int_Z \log\left(\frac{p(X, Z|\Theta)}{q(Z)}\right) q(Z) dZ \\
&= \text{ELBO}(\Theta, q)
\end{aligned} \tag{14}$$

4.1 Maximization of ELBO

When applying E-M as a Maximization-Maximization algorithm, we only need to optimize ELBO, let's revisit the ELBO-KL decomposition again:

$$\begin{aligned}
\mathcal{L}(\Theta|X) &= \int_Z \log\left(\frac{p(X, Z|\Theta)}{q(Z)}\right) q(Z) dZ + \int_Z \log\left(\frac{q(Z)}{p(Z|X, \Theta)}\right) q(Z) dZ \\
&= \text{ELBO}(\Theta, q) + \text{KL}(q(Z)||p(Z|X, \Theta))
\end{aligned} \tag{15}$$

STEP 1: fix $\Theta = \Theta^{(g)}$, maximize $q(Z)$ for ELBO

$$\begin{aligned}
q^*(\cdot) &= \arg \max_q \{\text{ELBO}(\Theta^{(g)}, q)\} \\
&= \arg \max_q \left\{ \int_Z \log\left(\frac{p(X, Z|\Theta^{(g)})}{q(Z)}\right) q(Z) dZ \right\} \\
&= p(Z|X, \Theta^{(g)}) \\
\implies \text{ELBO}(\Theta^{(g)}, q^*) &= \mathcal{L}(\Theta^{(g)}|X)
\end{aligned} \tag{16}$$

This follows from the realization that when fixing $\Theta = \Theta^{(g)}$, the ELBO is upper-bounded by $\mathcal{L}(\Theta^{(g)}|X)$, and the maximum is attained exactly when $\text{KL}(q^*||p(Z|X, \Theta^{(g)})) = 0$, i.e., $q^* = p(Z|X, \Theta^{(g)})$.

STEP 1 is invisible to the algorithm, but needs to be considered when interpreting convergence.

STEP 2: fix $q(\cdot) = p(Z|X, \Theta^{(g)})$, maximize Θ

substitute $q^* = p(Z|X, \Theta^{(g)})$ back into the ELBO:

$$\begin{aligned}\mathcal{L}(\Theta^{(g)}|X) &= \text{ELBO}(\Theta^{(g)}, q^*) \\ &= \int_Z \log \left(\frac{p(X, Z|\Theta^{(g)})}{p(Z|X, \Theta^{(g)})} \right) p(Z|X, \Theta^{(g)}) dZ \\ &= \underbrace{\int_Z \log(p(X, Z|\Theta^{(g)})) p(Z|X, \Theta^{(g)}) dZ}_{Q(\Theta^{(g)}|\Theta^{(g)})} - \underbrace{\int_Z \log(p(Z|X, \Theta^{(g)})) p(Z|X, \Theta^{(g)}) dZ}_{H(\Theta^{(g)}|\Theta^{(g)})}\end{aligned}\quad (17)$$

It is obvious that, after computing the E-M update from the decomposition in Eq.(17), we have:

$$\begin{aligned}\Theta^{(g+1)} &= \arg \max_{\Theta} Q(\Theta|\Theta^{(g)}) \\ &= \arg \max_{\Theta} \left(\int_Z \log(p(X, Z|\Theta)) p(Z|X, \Theta^{(g)}) dZ \right)\end{aligned}\quad (18)$$

then:

$$\underbrace{\int_Z \log(p(X, Z|\Theta^{(g+1)})) p(Z|X, \Theta^{(g)}) dZ}_{Q(\Theta^{(g+1)}|\Theta^{(g)})} \geq \underbrace{\int_Z \log(p(X, Z|\Theta^{(g)})) p(Z|X, \Theta^{(g)}) dZ}_{Q(\Theta^{(g)}|\Theta^{(g)})}\quad (19)$$

but if we can also prove:

$$\begin{aligned}H(\Theta|\Theta^{(g)}) &\geq H(\Theta^{(g)}|\Theta^{(g)}) \quad \forall \Theta \\ \implies H(\Theta^{(g+1)}|\Theta^{(g)}) &\geq H(\Theta^{(g)}|\Theta^{(g)})\end{aligned}\quad (20)$$

then we can show:

$$\begin{aligned}\mathcal{L}(\Theta^{(g+1)}|X) &= Q(\Theta^{(g+1)}|\Theta^{(g)}) + H(\Theta^{(g+1)}|\Theta^{(g)}) \\ &\geq \underbrace{Q(\Theta^{(g)}|\Theta^{(g)})}_{\text{Eq.(18)}} + \underbrace{H(\Theta^{(g)}|\Theta^{(g)})}_{\text{Eq.(20)}} \\ &= \mathcal{L}(\Theta^{(g)}|X)\end{aligned}\quad (21)$$

Note that since $Q(\Theta|\Theta^{(g)}) + H(\Theta|\Theta^{(g)}) = \mathcal{L}(\Theta|X)$ identically, jointly maximizing both terms would recover the (intractable) MLE itself. E-M maximizes only Q , so in general:

$$\begin{aligned}\bar{\Theta} &= \arg \max_{\Theta} \{Q(\Theta|\Theta^{(g)}) + H(\Theta|\Theta^{(g)})\} \\ \implies \mathcal{L}(\Theta^{(g+1)}) &\leq \mathcal{L}(\bar{\Theta})\end{aligned}\quad (22)$$

4.1.1 why $H(\Theta|\Theta^{(g)}) \geq H(\Theta^{(g)}|\Theta^{(g)}) \quad \forall \Theta$

1. cross entropy

$$\begin{aligned}
 H(\Theta|\Theta^{(g)}) &= - \int_Z \log(p(Z|X, \Theta)) p(Z|X, \Theta^{(g)}) dZ \\
 &= \text{Cross-Entropy}(p(Z|X, \Theta^{(g)}) \| p(Z|X, \Theta)) \\
 \implies \arg \min_{\Theta} \{H(\Theta|\Theta^{(g)})\} &= \Theta^{(g)} \\
 \implies \min_{\Theta} \{H(\Theta|\Theta^{(g)})\} &= \text{Entropy}(p(Z|X, \Theta^{(g)}))
 \end{aligned} \tag{23}$$

2. directly

$$\begin{aligned}
 &H(\Theta|\Theta^{(g)}) - H(\Theta^{(g)}|\Theta^{(g)}) \\
 &= \int_Z -\log(p(Z|X, \Theta)) p(Z|X, \Theta^{(g)}) dZ - \int_Z -\log(p(Z|X, \Theta^{(g)})) p(Z|X, \Theta^{(g)}) dZ \\
 &= \int_Z \log\left(\frac{p(Z|X, \Theta^{(g)})}{p(Z|X, \Theta)}\right) p(Z|X, \Theta^{(g)}) dZ \\
 &= \int_Z -\log\left(\frac{p(Z|X, \Theta)}{p(Z|X, \Theta^{(g)})}\right) p(Z|X, \Theta^{(g)}) dZ \\
 &\geq -\log\left(\int_Z \frac{p(Z|X, \Theta)}{p(Z|X, \Theta^{(g)})} p(Z|X, \Theta^{(g)}) dZ\right) \\
 &= 0 \quad \because \phi = -\log \text{ is a convex function}
 \end{aligned} \tag{24}$$

4.2 Proof of convergence without ELBO

let's decompose $\mathcal{L}(\Theta|X)$ into:

$$\begin{aligned}
 \mathcal{L}(\Theta|X) &= \log(p(X|\Theta)) \\
 &= \log(p(Z, X|\Theta)) - \log(p(Z|X, \Theta))
 \end{aligned} \tag{25}$$

take the expectation of both sides with respect to $p(Z|X, \Theta^{(g)})$:

$$\begin{aligned}
 \mathcal{L}(\Theta|X) &= \underbrace{\int_Z \log(p(Z, X|\Theta)) p(Z|X, \Theta^{(g)}) dZ}_{Q(\Theta|\Theta^{(g)})} - \underbrace{\int_Z \log(p(Z|X, \Theta)) p(Z|X, \Theta^{(g)}) dZ}_{H(\Theta|\Theta^{(g)})} \\
 \implies \mathcal{L}(\Theta^{(g)}|X) &= \underbrace{\int_Z \log(p(Z, X|\Theta^{(g)})) p(Z|X, \Theta^{(g)}) dZ}_{Q(\Theta^{(g)}|\Theta^{(g)})} - \underbrace{\int_Z \log(p(Z|X, \Theta^{(g)})) p(Z|X, \Theta^{(g)}) dZ}_{H(\Theta^{(g)}|\Theta^{(g)})}
 \end{aligned} \tag{26}$$

where $H(\Theta|\Theta^{(g)}) = \text{Cross-Entropy}(p(Z|X, \Theta^{(g)}) \| p(Z|X, \Theta))$.

Eq.(26) can be considered as a simpler version of the ELBO-KL decomposition of $\mathcal{L}(\Theta|X)$ we have seen previously in Eq.(17).

5 E-M Example: Gaussian Mixture Model

Gaussian Mixture Model (k-mixture) (GMM):

$$p(x|\Theta) = \sum_{l=1}^k \alpha_l \mathcal{N}(x|\mu_l, \Sigma_l) \quad \sum_{l=1}^k \alpha_l = 1 \quad (27)$$

and $\Theta = \{\alpha_1, \dots, \alpha_k, \mu_1, \dots, \mu_k, \Sigma_1, \dots, \Sigma_k\}$

For data $X = \{x_1, \dots, x_n\}$ we introduce “latent” variables $Z = \{z_1, \dots, z_n\}$, where each z_i indicates which mixture component x_i belongs to.

Looking at the E-M algorithm:

$$\Theta^{(g+1)} = \arg \max_{\Theta} [q(\Theta, \Theta^{(g)})] = \arg \max_{\Theta} \left(\int_Z \log(p(X, Z|\Theta)) p(Z|X, \Theta^{(g)}) dZ \right) \quad (28)$$

All we need to do is to define both $p(X, Z|\Theta)$ and $p(Z|X, \Theta)$

5.1 Gaussian Mixture Model in action

$$p(X|\Theta) = \prod_{i=1}^n p(x_i|\Theta) = \prod_{i=1}^n \sum_{l=1}^k \alpha_l \mathcal{N}(x_i|\mu_l, \Sigma_l) \quad (29)$$

How to define $p(X, Z|\Theta)$

$$p(X, Z|\Theta) = \prod_{i=1}^n p(x_i, z_i|\Theta) = \prod_{i=1}^n \underbrace{p(x_i|z_i, \Theta)}_{\mathcal{N}(x_i|\mu_{z_i}, \Sigma_{z_i})} \underbrace{p(z_i|\Theta)}_{\alpha_{z_i}} = \prod_{i=1}^n \alpha_{z_i} \mathcal{N}(x_i|\mu_{z_i}, \Sigma_{z_i}) \quad (30)$$

Notice that $p(X, Z|\Theta)$ is actually simpler than $p(X|\Theta)$.

How to define $p(Z|X, \Theta)$

$$p(Z|X, \Theta) = \prod_{i=1}^n p(z_i|x_i, \Theta) = \prod_{i=1}^n \frac{\alpha_{z_i} \mathcal{N}(x_i|\mu_{z_i}, \Sigma_{z_i})}{\sum_{l=1}^k \alpha_l \mathcal{N}(x_i|\mu_l, \Sigma_l)} \quad (31)$$

5.2 The E-Step

$$\begin{aligned} q(\Theta, \Theta^{(g)}) &= \int_Z \log(p(X, Z|\Theta)) p(Z|X, \Theta^{(g)}) dZ \\ &= \int_{z_1} \cdots \int_{z_n} \left(\sum_{i=1}^n \log p(z_i, x_i|\Theta) \prod_{i=1}^n p(z_i|x_i, \Theta^{(g)}) \right) dz_1 \cdots dz_n \end{aligned} \quad (32)$$

5.3 Some derivation to help

let $p(Y)$ be the joint pdf $p(y_1, \dots, y_N)$, and let $F(Y)$ be an additively separable function, where each term involves only one variable y_i , i.e.,

$$F(Y) = f_1(y_1) + \cdots + f_N(y_N) = \sum_{i=1}^N f_i(y_i) \quad (33)$$

then,

$$\int_{y_1} \cdots \int_{y_N} \left(\sum_{i=1}^N f_i(y_i) \right) p(Y) dY = \sum_{i=1}^N \left(\int_{y_i} f_i(y_i) p(y_i) dy_i \right) \quad (34)$$

5.3.1 Proof

$$\int_Y F(Y)p(Y)dY = \int_{y_1} \int_{y_2} \cdots \int_{y_N} \left(\sum_{i=1}^N f_i(y_i) \right) p(Y) dy_1 \cdots dy_N \quad (35)$$

Expanding it out, this equation has N sum terms:

$$\begin{aligned} &= \int_{y_1} \int_{y_2} \cdots \int_{y_N} f_1(y_1) p(y_1, \dots, y_N) \prod_{i=1}^N (dy_i) + \cdots + \int_{y_1} \int_{y_2} \cdots \int_{y_N} f_N(y_N) p(y_1, \dots, y_N) \prod_{i=1}^N (dy_i) \\ &= \int_{y_1} f_1(y_1) dy_1 \left(\int_{y_2} \cdots \int_{y_N} p(y_1, \dots, y_N) \prod_{i=2}^N (dy_i) \right) + \cdots + \int_{y_N} f_N(y_N) dy_N \int_{y_1} \cdots \int_{y_{N-1}} p(y_1, \dots, y_N) \prod_{i=1}^{N-1} (dy_i) \end{aligned} \quad (36)$$

the inner integral in the first big bracket is the marginal probability density $p(y_1)$; therefore, the first term becomes:

$$\int_{y_1} f_1(y_1) p(y_1) dy_1 \quad (37)$$

Applying this to each of the N terms, therefore:

$$\int_Y F(Y)p(Y)dY = \int_{y_1} f_1(y_1) p(y_1) dy_1 + \cdots + \int_{y_N} f_N(y_N) p(y_N) dy_N \quad (38)$$

now apply Eq.(34), we have:

$$\begin{aligned} q(\Theta, \Theta^{(g)}) &= \int_{z_1} \cdots \int_{z_n} \left(\sum_{i=1}^n \log p(z_i, x_i | \Theta) \prod_{i=1}^n p(z_i | x_i, \Theta^{(g)}) \right) dz_1 \cdots dz_n \\ &= \sum_{i=1}^n \left(\int_{z_i} \log p(z_i, x_i | \Theta) p(z_i | x_i, \Theta^{(g)}) dz_i \right) \quad z_i \in \{1, \dots, k\} \\ &= \sum_{l=1}^k \sum_{i=1}^n \log p(z_i = l, x_i | \Theta) p(z_i = l | x_i, \Theta^{(g)}) \quad \text{swap the summation order} \\ &= \sum_{l=1}^k \sum_{i=1}^n \log[\alpha_l \mathcal{N}(x_i | \mu_l, \Sigma_l)] p(l | x_i, \Theta^{(g)}) \quad \text{substitute the Gaussian, write } z_i = l \text{ as } l \end{aligned} \quad (39)$$

5.4 The M-Step objective function

$$\begin{aligned} q(\Theta, \Theta^{(g)}) &= \sum_{l=1}^k \sum_{i=1}^n \log[\alpha_l \mathcal{N}(x_i | \mu_l, \Sigma_l)] p(l | x_i, \Theta^{(g)}) \\ &= \sum_{l=1}^k \sum_{i=1}^n \log(\alpha_l) p(l | x_i, \Theta^{(g)}) + \sum_{l=1}^k \sum_{i=1}^n \log[\mathcal{N}(x_i | \mu_l, \Sigma_l)] p(l | x_i, \Theta^{(g)}) \end{aligned} \quad (40)$$

5.4.1 computing responsibilities

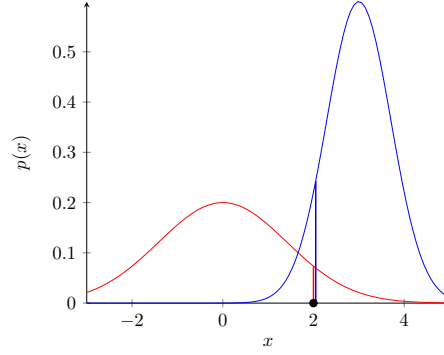


Figure 3: computing responsibility probabilities; red and blue each indicate their own unnormalized responsibilities

$$p(l|x_i, \Theta^{(g)}) = \frac{\alpha_l \mathcal{N}(\mathbf{x}_i | \mu_l, \Sigma_l)}{\sum_{s=1}^k \alpha_s \mathcal{N}(\mathbf{x}_i | \mu_s, \Sigma_s)} \quad (41)$$

Eq.(40) shows that the first term contains only α and the second term contains only μ, Σ . So we can maximize both terms independently.

5.5 The M-Step: maximizing α

Maximizing α means that:

$$\frac{\partial \sum_{l=1}^k \sum_{i=1}^n \log(\alpha_l) p(l|x_i, \Theta^{(g)})}{\partial \alpha_1, \dots, \partial \alpha_k} = [0 \dots 0] \quad \text{subject to } \sum_{l=1}^k \alpha_l = 1 \quad (42)$$

This is to be solved using Lagrange Multiplier

$$\mathbb{LM}(\alpha_1, \dots, \alpha_k, \lambda) = \sum_{l=1}^k \log(\alpha_l) \underbrace{\left(\sum_{i=1}^n p(l|x_i, \Theta^{(g)}) \right)}_{\text{contains no } \alpha} + \lambda \left(\sum_{l=1}^k \alpha_l - 1 \right) \quad (43)$$

taking derivative with respect to just one α_l :

$$\begin{aligned} \frac{\partial \mathbb{LM}}{\partial \alpha_l} &= \frac{1}{\alpha_l} \left(\sum_{i=1}^n p(l|x_i, \Theta^{(g)}) \right) + \lambda = 0 \\ \implies -\lambda &= \frac{1}{\alpha_l} \left(\sum_{i=1}^n p(l|x_i, \Theta^{(g)}) \right) \\ \implies -\lambda \alpha_l &= \left(\sum_{i=1}^n p(l|x_i, \Theta^{(g)}) \right) \\ \implies -\lambda \sum_{l=1}^k \alpha_l &= \sum_{l=1}^k \left(\sum_{i=1}^n p(l|x_i, \Theta^{(g)}) \right) \\ \implies \lambda &= -n \end{aligned} \quad (44)$$

substitute $\lambda = -n$ into Eq.(44):

$$\begin{aligned}
& \frac{1}{\alpha_l} \left(\sum_{i=1}^n p(l|x_i, \Theta^{(g)}) \right) + \lambda = 0 \\
\Rightarrow & \frac{1}{\alpha_l} \left(\sum_{i=1}^n p(l|x_i, \Theta^{(g)}) \right) - n = 0 \\
& \Rightarrow \alpha_l = \frac{1}{n} \sum_{i=1}^n p(l|x_i, \Theta^{(g)})
\end{aligned} \tag{45}$$

5.6 The M-Step: maximizing μ, Σ

Here I jot down the MLE steps for the parameters of a single multidimensional Gaussian distribution. The MLE of a single 1D Gaussian distribution can be easily found in your previous work.

So, maximizing μ, Σ means that:

$$\frac{\partial \sum_{l=1}^k \sum_{i=1}^n \log[\mathcal{N}(x_i|\mu_l, \Sigma_l)] p(l|x_i, \Theta^{(g)})}{\partial \mu_1, \dots, \partial \mu_k, \partial \Sigma_1, \dots, \partial \Sigma_k} = [0 \dots 0] \tag{46}$$

You will need some linear algebra identities to solve this; it is quite involved. For full details, refer to Bilmes' gentle tutorial [5].

5.6.1 Some formulas to remember

- derivatives of log of determinant (**with** determinant)

$$\frac{\partial \log |\mathbf{X}|}{\partial \mathbf{X}} = (\mathbf{X}^{-1})^\top \tag{47}$$

- Derivatives of Traces

$$\frac{\partial \text{Tr}(F(\mathbf{X}))}{\partial \mathbf{X}} = (f(\mathbf{X}))^\top \tag{48}$$

where $f(\cdot)$ is the **scalar derivative** of $F(\cdot)$

- Derivatives of Traces, fact 1

$$\frac{\partial \text{Tr}(\mathbf{A}\mathbf{X}\mathbf{B})}{\partial \mathbf{X}} = \mathbf{A}^\top \mathbf{B}^\top \tag{49}$$

- Derivatives of Traces of inverse, fact 2

$$\frac{\partial \text{Tr}((\mathbf{X} + \mathbf{A})^{-1})}{\partial \mathbf{X}} = -((\mathbf{X} + \mathbf{A})^{-1}(\mathbf{X} + \mathbf{A})^{-1})^\top \tag{50}$$

- Derivatives of Traces of inverse, fact 3

$$\frac{\partial \text{Tr}(\mathbf{A}\mathbf{X}^{-1}\mathbf{B})}{\partial \mathbf{X}} = -(\mathbf{X}^{-1}\mathbf{B}\mathbf{A}\mathbf{X}^{-1})^\top \tag{51}$$

5.6.2 Maximization μ_l

$$\begin{aligned}
\text{second part of } q(\Theta, \Theta^{(g)}) &= \sum_{l=1}^k \sum_{i=1}^n \log[\mathcal{N}(x_i | \mu_l, \Sigma_l)] p(l | x_i, \Theta^{(g)}) \\
&= \sum_{i=1}^n \sum_{l=1}^k \log \left(\frac{1}{\sqrt{(2\pi)^d |\Sigma_l|}} \exp \left(-\frac{1}{2} (x_i - \mu_l)^\top \Sigma_l^{-1} (x_i - \mu_l) \right) \right) p(l | x_i, \Theta^{(g)})
\end{aligned} \tag{52}$$

As a template, recall the log-likelihood of N i.i.d. zero-meaned columns $\mathbf{Y}$ (each column in $\mathbb{R}^D$) under $\mathcal{N}(\mathbf{0}, \mathcal{K})$:

$$\mathcal{L} \equiv \mathcal{L}(p(\mathbf{Y} | \mathcal{K})) = -\frac{DN}{2} \log(2\pi) - \frac{N}{2} \log |\mathcal{K}| - \frac{1}{2} \text{Tr}(\mathcal{K}^{-1} \mathbf{Y} \mathbf{Y}^\top) \tag{53}$$

Let $\mathbf{Y}$ be the zero-meaned data matrix, where each **column** of $\mathbf{Y}$ is $\mathbf{x}_i - \mu_l$:

$$\mathcal{S}(\mu_l, \Sigma_l) = \sum_{i=1}^n -\frac{1}{2} \log(|\Sigma_l|) p(l | x_i, \Theta^{(g)}) - \sum_{i=1}^n \frac{1}{2} (x_i - \mu_l)^\top \Sigma_l^{-1} (x_i - \mu_l) p(l | x_i, \Theta^{(g)}) + \text{Const.} \tag{54}$$

$$\begin{aligned}
\Rightarrow \mathcal{S}(\mu_l, \Sigma_l^{-1}) &= -\text{Tr} \left(\frac{\Sigma_l^{-1}}{2} \sum_{i=1}^n (x_i - \mu_l)(x_i - \mu_l)^\top p(l | x_i, \Theta^{(g)}) \right) + \text{Constant} \\
\Rightarrow \frac{\partial \mathcal{S}(\mu_l, \Sigma_l^{-1})}{\partial \mu_l} &= \frac{\Sigma_l^{-1}}{2} \sum_{i=1}^n 2(x_i - \mu_l) p(l | x_i, \Theta^{(g)}) = 0 \\
\Rightarrow \sum_{i=1}^n x_i p(l | x_i, \Theta^{(g)}) &= \mu_l \sum_{i=1}^n p(l | x_i, \Theta^{(g)}) \\
\Rightarrow \mu_l &= \frac{\sum_{i=1}^n x_i p(l | x_i, \Theta^{(g)})}{\sum_{i=1}^n p(l | x_i, \Theta^{(g)})}
\end{aligned} \tag{55}$$

5.6.3 Maximization of covariance

$$\begin{aligned}
\text{second part of } q(\Theta, \Theta^{(g)}) &= \sum_{l=1}^k \sum_{i=1}^n \log[\mathcal{N}(x_i | \mu_l, \Sigma_l)] p(l | x_i, \Theta^{(g)}) \\
&= \sum_{i=1}^n \sum_{l=1}^k \log \left(\frac{1}{\sqrt{(2\pi)^d |\Sigma_l|}} \exp \left(-\frac{1}{2} (x_i - \mu_l)^\top \Sigma_l^{-1} (x_i - \mu_l) \right) \right) p(l | x_i, \Theta^{(g)})
\end{aligned} \tag{56}$$

- let $\mathbf{Y}$ be zero-meaned data matrix, where each **column** of $\mathbf{Y}$ is $x_i - \mu_l$
- let $\mathbf{P}$ be diagonal matrix in which $\mathbf{P}_{ii}$ correspond to $p(l | x_i, \Theta^{(g)})$

$$\mathcal{L} \equiv \mathcal{L}(p(\mathbf{Y} | \mu_l, \Sigma_l)) = -\frac{d \times \text{Tr}(\mathbf{P})}{2} \log(2\pi) - \frac{\text{Tr}(\mathbf{P})}{2} \log |\Sigma_l| - \frac{1}{2} \text{Tr}(\Sigma_l^{-1} \mathbf{Y} \mathbf{P} \mathbf{Y}^\top) \tag{57}$$

$$\begin{aligned}
\frac{\partial \mathcal{L}}{\partial \Sigma_l} &= \Sigma_l^{-1} \mathbf{Y} \mathbf{P} \mathbf{Y}^\top \Sigma_l^{-1} - \text{Tr}(\mathbf{P}) \Sigma_l^{-1} = \mathbf{0} \\
\implies \Sigma_l^{-1} \mathbf{Y} \mathbf{P} \mathbf{Y}^\top \Sigma_l^{-1} &= \text{Tr}(\mathbf{P}) \Sigma_l^{-1} \\
\implies \mathbf{Y} \mathbf{P} \mathbf{Y}^\top \Sigma_l^{-1} &= \text{Tr}(\mathbf{P}) \mathbf{I} \\
\implies \Sigma_l^{-1} &= \text{Tr}(\mathbf{P}) (\mathbf{Y} \mathbf{P} \mathbf{Y}^\top)^{-1} \\
\implies \Sigma_l &= \text{Tr}(\mathbf{P})^{-1} (\mathbf{Y} \mathbf{P} \mathbf{Y}^\top) = \frac{(\mathbf{Y} \mathbf{P} \mathbf{Y}^\top)}{\text{Tr}(\mathbf{P})} \\
&= \frac{\sum_{i=1}^n (x_i - \mu_l)(x_i - \mu_l)^\top p(l|x_i, \Theta^{(g)})}{\sum_{i=1}^n p(l|x_i, \Theta^{(g)})}
\end{aligned} \tag{58}$$

$$\mathcal{S}(\mu_l, \Sigma_l^{-1}) = \sum_{i=1}^n \left(-\frac{1}{2} \log(|\Sigma_l|) - \frac{1}{2} (x_i - \mu_l)^\top \Sigma_l^{-1} (x_i - \mu_l) \right) p(l|x_i, \Theta^{(g)}) \tag{59}$$

Change Σ to Σ^{-1} , this is so that after taking derivative of $\log(X)$, the result is in terms of X^{-1}

$$\begin{aligned}
&= \left(\frac{1}{2} \sum_{i=1}^n \log(|\Sigma_l^{-1}|) p(l|x_i, \Theta^{(g)}) - \frac{1}{2} \text{tr} \left(\Sigma_l^{-1} \underbrace{\sum_{i=1}^n (x_i - \mu_l)(x_i - \mu_l)^\top p(l|x_i, \Theta^{(g)})}_{M_l} \right) \right) \\
\implies \frac{\partial \mathcal{S}(\mu_l, \Sigma_l^{-1})}{\partial \Sigma_l^{-1}} &= \frac{2 \sum_{i=1}^n \Sigma_l p(l|x_i, \Theta^{(g)}) - \sum_{i=1}^n \text{diag}(\Sigma_l) p(l|x_i, \Theta^{(g)})}{2} - \frac{2M_l - \text{diag}(M_l)}{2} = 0 \\
\implies 2 \left(\sum_{i=1}^n \Sigma_l p(l|x_i, \Theta^{(g)}) - M_l \right) - \text{diag} \left(\sum_{i=1}^n \Sigma_l p(l|x_i, \Theta^{(g)}) - M_l \right) &= 0 \\
\implies \sum_{i=1}^n \Sigma_l p(l|x_i, \Theta^{(g)}) - M_l &= 0 \\
\implies \Sigma_l &= \frac{M_l}{\sum_{i=1}^n p(l|x_i, \Theta^{(g)})} = \frac{\sum_{i=1}^n (x_i - \mu_l)(x_i - \mu_l)^\top p(l|x_i, \Theta^{(g)})}{\sum_{i=1}^n p(l|x_i, \Theta^{(g)})}
\end{aligned} \tag{60}$$

5.7 Summary of Gaussian Mixture Model

In summary, to update $\Theta^{(g)} \rightarrow \Theta^{(g+1)}$:

$$\alpha_l^{(g+1)} = \frac{1}{n} \sum_{i=1}^n p(l|x_i, \Theta^{(g)}) \tag{61}$$

$$\mu_l^{(g+1)} = \frac{\sum_{i=1}^n x_i p(l|x_i, \Theta^{(g)})}{\sum_{i=1}^n p(l|x_i, \Theta^{(g)})} \tag{62}$$

$$\Sigma_l^{(g+1)} = \frac{\sum_{i=1}^n [x_i - \mu_l^{(g+1)}][x_i - \mu_l^{(g+1)}]^\top p(l|x_i, \Theta^{(g)})}{\sum_{i=1}^n p(l|x_i, \Theta^{(g)})} \tag{63}$$

One can verify that when the number of components $k = 1 \implies p(l = 1|x_i, \Theta^{(g)}) = 1$, then:

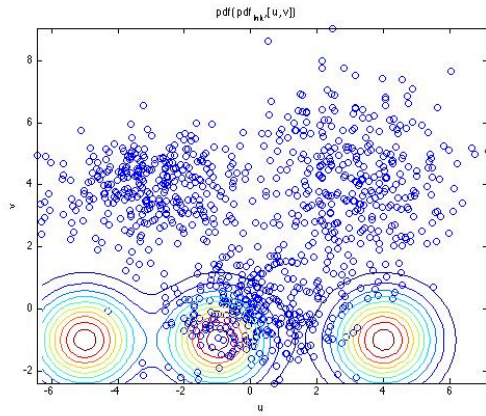
$$\alpha_l^{(g+1)} = 1 \tag{64}$$

$$\mu_l^{(g+1)} = \frac{1}{n} \sum_{i=1}^n x_i \quad (65)$$

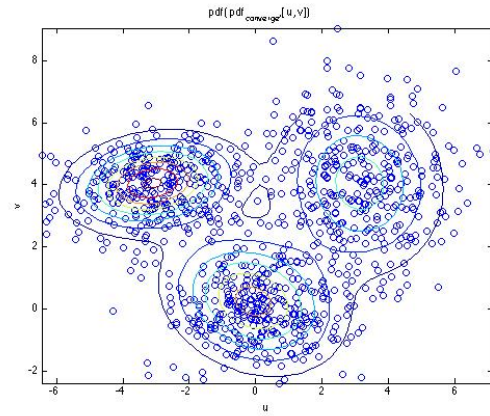
$$\Sigma_l^{(g+1)} = \frac{1}{n} \sum_{i=1}^n [x_i - \mu_l^{(g+1)}][x_i - \mu_l^{(g+1)}]^\top \quad (66)$$

which becomes the parameter update of a single Gaussian

5.8 GMM fitting: initialization vs. convergence



(a) GMM at initialization: $\Theta^{(1)}$



(b) GMM at convergence: $\Theta^{(\text{final})}$

6 Exercises

1. **(Jensen's inequality.)** Let $\phi(x) = -\log(x)$ and let $p_1 = p_2 = \frac{1}{2}$, $x_1 = 1$, $x_2 = 4$. Compute both sides of Jensen's inequality $\phi(\sum_i p_i x_i) \leq \sum_i p_i \phi(x_i)$ and verify the inequality holds. For which x_1, x_2 does equality hold?
2. **(Responsibilities.)** A 1-D GMM has two components: $\alpha_1 = 0.4$, $\mathcal{N}(x | 0, 2)$ and $\alpha_2 = 0.6$, $\mathcal{N}(x | 3, 0.5)$. For the observation $x = 2$, compute the responsibilities $p(l | x, \Theta)$ for $l = 1, 2$. (Three decimal places.)
3. **(The α update.)** Following the Lagrange-multiplier derivation of Section 5, show that for a $k = 2$ mixture with $n = 4$ data points whose responsibilities to component 1 are 0.9, 0.8, 0.2, 0.1, the update gives $\alpha_1^{(g+1)} = \frac{1}{2}$. Verify $\alpha_1^{(g+1)} + \alpha_2^{(g+1)} = 1$ without computing α_2 from its own responsibilities.
4. **(Single-Gaussian sanity check.)** Set $k = 1$ in the three M-step update formulas of Section 5 and show each reduces to the ordinary maximum-likelihood estimate of a single Gaussian.
5. **(Why H cannot decrease.)** Using Jensen's inequality with the convex function $\phi = -\log$, prove that $H(\Theta | \Theta^{(g)}) - H(\Theta^{(g)} | \Theta^{(g)}) \geq 0$ for all Θ , and state exactly where the proof uses $\int_Z p(Z | X, \Theta) dZ = 1$.

7 Solutions

1. LHS: $\phi(\frac{1}{2} \cdot 1 + \frac{1}{2} \cdot 4) = -\log(2.5) \approx -0.916$. RHS: $\frac{1}{2}(-\log 1) + \frac{1}{2}(-\log 4) = -\frac{1}{2} \log 4 \approx -0.693$. Indeed $-0.916 \leq -0.693$. Equality holds iff $x_1 = x_2$ (the chord degenerates to a point; $-\log$ is strictly convex).
2. Unnormalized weights: $w_1 = 0.4 \cdot \frac{1}{\sqrt{2\pi \cdot 2}} e^{-\frac{(2-0)^2}{2 \cdot 2}} = 0.4 \cdot 0.2820 \cdot e^{-1} \approx 0.0415$; $w_2 = 0.6 \cdot \frac{1}{\sqrt{2\pi \cdot 0.5}} e^{-\frac{(2-3)^2}{2 \cdot 0.5}} = 0.6 \cdot 0.5642 \cdot e^{-1} \approx 0.1245$. Normalizing: $p(1 | x) \approx 0.250$, $p(2 | x) \approx 0.750$.
3. $\alpha_1^{(g+1)} = \frac{1}{n} \sum_i p(1 | x_i, \Theta^{(g)}) = \frac{1}{4}(0.9 + 0.8 + 0.2 + 0.1) = \frac{2}{4} = \frac{1}{2}$. Since responsibilities per point sum to one, $\sum_i p(2 | x_i) = n - \sum_i p(1 | x_i) = 2$, so $\alpha_2^{(g+1)} = \frac{2}{4} = \frac{1}{2}$ and the two sum to one automatically.
4. With $k = 1$, $p(l=1 | x_i, \Theta^{(g)}) = 1$ for every i . Then $\alpha_1 = \frac{1}{n} \sum_i 1 = 1$; $\mu_1 = \frac{\sum_i x_i}{n}$ (sample mean); $\Sigma_1 = \frac{1}{n} \sum_i (x_i - \mu_1)(x_i - \mu_1)^\top$ (sample covariance) — the single-Gaussian MLE.
5. $H(\Theta | \Theta^{(g)}) - H(\Theta^{(g)} | \Theta^{(g)}) = \int_Z -\log\left(\frac{p(Z|X,\Theta)}{p(Z|X,\Theta^{(g)})}\right) p(Z | X, \Theta^{(g)}) dZ \geq -\log \int_Z \frac{p(Z|X,\Theta)}{p(Z|X,\Theta^{(g)})} p(Z | X, \Theta^{(g)}) dZ$ by Jensen ($\phi = -\log$ convex, expectation under $p(Z | X, \Theta^{(g)})$). The inner integral collapses to $\int_Z p(Z | X, \Theta) dZ = 1$ — used exactly here — giving $-\log 1 = 0$.

References

- [1] Christopher M Bishop, *Pattern Recognition and Machine Learning*, Springer, 2006.
- [2] Arthur P. Dempster, Nan M. Laird, and Donald B. Rubin, "Maximum likelihood from incomplete data via the EM algorithm," *Journal of the Royal Statistical Society: Series B*, vol. 39, no. 1, pp. 1–38, 1977.
- [3] C. F. Jeff Wu, "On the convergence properties of the EM algorithm," *The Annals of Statistics*, vol. 11, no. 1, pp. 95–103, 1983.
- [4] Radford M. Neal and Geoffrey E. Hinton, "A view of the EM algorithm that justifies incremental, sparse, and other variants," in *Learning in Graphical Models*, Michael I. Jordan, Ed., pp. 355–368. Kluwer Academic Publishers, 1998.
- [5] Jeff A. Bilmes, "A gentle tutorial of the EM algorithm and its application to parameter estimation for Gaussian mixture and hidden Markov models," Tech. Rep. TR-97-021, International Computer Science Institute, Berkeley, 1998.